



No. DGMS(Tech.)/ S&T/Circular (Approval) No. Dhanbad, dated 14.08.2026

To

All Mine Owners, Agents and Managers;
All Manufacturers of e-Exploders & e-Detonators;
All scientific and other institutions related to testing of e-Exploders & e-Detonators.

Subject: Guidelines for Testing Standards of permitted e-Exploder for use in underground mines.

Sir,

1. The Ministry of Commerce and Industry, Department for Promotion of Industry and Internal Trade (DPIIT) had issued Gazette Notification No. S.O. 4255(E) dated 29th September 2023, regarding the phased discontinuation of electric detonators concerning the issues of public safety and National security. The government's assessment concluded the need for complete prohibition of electric detonator by the end of FY 2024-25.
2. As a result, the blasting technology in the mining industry depended on the new development i.e. Electronic Detonator (e-Detonator) and Electronic Exploder (e-Exploder) system. As there was no guidelines/standards for construction, testing and use of e-Exploder. This resulted into the situation that no approved e-Exploder was available to fire the e-Detonators in underground coal mines which may stop the coal production from underground mines. Although, standards for e-detonator were developed and notified as IS 18462:2023, however few manufacturers developed the products [e-Detonator & e-Exploder].
3. Keeping in view of this situation, the Directorate General of Mines Safety (DGMS) constituted a committee to prepare Guidelines for e-Exploder, particularly to be used in coal mines and underground metalliferous mines (gassy, where Regulation 142 of the Metalliferous Mines Regulations, 1961 is applicable) and other hazardous locations. The Committee consisted of members from the DGMS, CSIR-CIMFR, IIT-ISM, CIL, CMPDIL and representatives from explosive and accessories manufacturers (M/s SOLAR, M/s CDET, M/s IDEAL, M/s Regenesi Industries Pvt. Ltd. and M/s Kakatiya Industries).
4. The detailed deliberations were made by the committee in physical and Hybrid mode. The draft e-Exploder Guidelines were finally prepared and circulated to all concerned on 16th July, 2024. On the basis of these guidelines, around 5 manufacturers (Solar,

C-Det, IDEAL, Regenesis and Kakatiya) manufactured the e-Exploders which were tested at CISR-CIMFR, Dhanbad and other Govt./NABL approved Labs.

5. In this regard, deliberations were also made in the National Workshop on "Latest Advancements in the Use of Explosives and Blasting Technology in Mines and the Way Forward" conducted at Ranchi on 14th & 15th November, 2025, in which approximately 250 participants, including manufacturers, scientists, academicians, product experts, and industry users, took part. Inputs of this workshop were also addressed in the guidelines.
6. Further, a meeting was convened on 06.04.2026 at the DGMS Headquarters, attended by representatives of equipment manufacturers, accredited test houses, academicians, scientists, and officials of DGMS. The meeting reviewed the testing protocol of the e- Exploder prepared by the Committee.
7. After detailed discussion and deliberations on the various aspect of test standards and passing out criteria of e-Exploders for use in underground coal mines and underground metalliferous mines (gassy, where Regulation 142 of the Metalliferous Mines Regulations, 1961 is applicable), a set of guidelines were framed which is given at **Annexure-I (Appendix-A, Appendix-B & Appendix-C)**.
8. In this connection, Department of Consumer Affairs, Bureau of Indian Standards has also issued "IS 19359: 2026 e-Exploders-Specification" regarding the testing standards required to be done for e-Exploder before approval and before placing them into use in mines, vide Gazette notification Ref: HQ-PUB013/1/2020-PUB-BIS (1546) New Delhi, the 27th May, 2026, enclosed as **Annexure-II**.
9. All user(s), manufacturers of e-Exploders and all scientific and other institutions related to testing of e-Exploders & e-Detonators should ensure through a mechanism established for the purpose that e-Exploders have complied with the relevant standards and guidelines before putting them into use in coal mines and underground metalliferous mines (gassy, where Regulation 142 of the Metalliferous Mines Regulations, 1961 is applicable).
10. All user(s) in consultation with respective manufacturers shall prepare a comprehensive SOP for storage, transport, handling, charging and firing of e-Detonator with e- Exploder (Electronic blasting system) and the same shall be strictly complied with.


(Ujjwal Tah) 14/08/26

**Director General of Mines Safety/
Chief Inspector-cum-Facilitator of Mines**

e-Exploder – (Guidelines)**FOREWORD**

An Electronic Exploder (e-Exploder) is a set of devices consisting of a Logger, a Blaster and a Continuity Tester as independent devices with distinct functions and with independent authorization to ensure safe use.

This device(s) may be used exclusively to communicate with electronic detonators. They usually come with a specific method and protocol of communication compatible to a specific electronic detonator built with the same method and protocol of communication structure and so shall not be interchangeable. In Electronic Detonator systems, an e-Exploder from one manufacturer shall not initiate Electronic Detonators from another manufacturer due to differing communication systems between the e-Exploder and Electronic Detonators.

The e-Exploder comes in a form easy to carry to the mine /field by an authorized shotfirer and contains a human interface option in the form of a key-board or other form with multiple operational options that can be used to issue different operational commands to complete the process of defining, programming, checking and finally firing of each batch of electronic detonators. e-Exploders come with construction features suitable to be used both in underground mines and in open cast mines.

The e-exploder shall facilitate operation of the blasting from a safe distance and e-Exploder meant for use in underground mines shall additionally come with intrinsically safe features as approved by Director General Mines Safety (DGMS).

The e-Exploder comes with core features to check the health status of the electronic detonator at preliminary stage, to load the serial number and delay time in each electronic detonator at next stage and to check and confirm the electrical connectivity of each detonator to the trunk line cable prior to the blasting phase; once moved to the blasting phase, the e-Exploder shall offer features like checking the attendance of all connected electronic detonators, to initiate charge loading and a possible verification of the same and then after moving to issue the final fire command with a provision for issuing a defuse/abort command to meet some uncommon eventualities.

The e-Exploder comes in a packing case easy to carry and robust enough to withstand the rigors in a mine environment and also equipped with a physical access key to turn on and turn off in addition to a password-based authentication for issue of commands.

1. SCOPE

1.1 This standard covers the requirements of e-Exploders and portable blasting/logging instruments used for shot firing operations for activating electronic detonator(s) during mining and non-mining blasting operations.

2. DEFINITIONS:

2.1 Electronic detonator: Detonator assembly in which the time delay is achieved by means of an electronic chip activated by electronic signature signal/ stimuli.

2.2 Electronic Exploder System [e-Exploder]: A set of devices which uses electrical energy and coded signal for programming/communicating/verifying/activating/initiating/ blasting of electronic detonator(s). It consists of a Logger, a Blaster and a Continuity Tester as independent devices with distinct functions and with independent authorization to ensure safe use.

2.3 Programmer: Also called as logger is an instrument used to program the serial number and the required delay into an electronic detonator with a mechanism to verify the same along with the healthiness of the detonator(s).

2.4 Blaster: Blaster is an instrument used to carry out the final operations in the blasting sequence involving verification of the connected detonators from the firing chamber, to arm or charge the detonators and to issue the fire/defuse command.

2.5 The continuity tester: Also known as connection tester is an instrument to verify the parallel connection of each detonator to the trunk line. The continuity tester used in underground mines will additionally come with intrinsically safe features as approved by Director General Mines Safety (DGMS).

3. CLASSIFICATIONS:

3.1 The e-Exploders covered by this standard shall be either of the following two types:

Permitted Type -A type which is approved by the statutory authority namely, Directorate General of Mines Safety, for use in underground coal mines, underground Metalliferous mines (gassy, where Regulation 142 of the Metalliferous Mines Regulations, 1961 is applicable) and other hazardous locations.

General or Non-permitted Type - A type which is used for mining operations in open places or civil work where the danger from flammable gases is not present.

4.0 CONSTRUCTION AND GENERAL REQUIREMENTS:

4.1 The e-Exploder shall be so constructed that all parts are of satisfactory strength for the service intended. The enclosure case shall be robust such that it meets the requirements of Clause No. 8.4 and 8.5, splash proof and dust tight (degree of protection IP 54) in accordance with IS/IEC 60529:2001* as amended time to time.

The enclosure of the permitted type e-Exploders shall be so constructed as not to give rise to ignition hazard of gas and coal dust due to frictional spark, static charges and leakage current.

The circuit of the e-Exploder shall be intrinsically safe conforming to IEC/IS: 60079-11-2023#, as amended time to time.

(*Specification for degrees of protection provided by enclosures for low-voltage switchgear and control gear.)

[# Explosive Atmospheres: Equipment Protection by Intrinsic Safety "ia"]

4.2 Carrying Strap/handle/Case- A carrying strap/handle shall be fastened securely to the body of the e-Exploder or its carrying case. The carrying strap shall be of adequate strength to withstand an additional 50 percent weight of e-Exploder for 15 minutes without failure or break.

4.3 The output terminals shall be provided with suitable connecting devices for firm connections. It should not be possible to remove the connecting device from its position. It shall be possible for easy and firm connection of shot firing cable with the terminals.

4.4 The firing circuit of the e-Exploder shall be adequately insulated from its enclosure and shall meet the requirements given in Clause No. 8.1.

4.5 It shall not be possible to keep the e-Exploder devices in "ON" condition unless the key is

inserted. The operating key shall be specific to each e-Exploder to avoid the operations by unauthorized persons using a generally available key of similar design or construction.

4.6 The e-Exploder (Blasting unit) shall be so constructed that:

- a) It shall be operated by a removable key followed by an authentication password for activation.
- b) The arming/fire command from the blaster of the e-Exploder shall be by an additional and separate password enabled operation.
- c) No residual energy is left at the terminals after the completion of operation sequence.
- d) The circuit of the blasting unit shall be intrinsically safe conforming to IEC/IS: 60079-11-2023, as amended time to time.

4.7 The permitted type e-Exploders shall have suitable devices to terminate the output energy pulse within **four milliseconds** of the release of the firing command.

4.8 Each e-Exploder shall be capable of firing a prescribed number of electronic detonators in the manner recommended by the manufacturer of the e-Exploder. The performance requirements of the e-Exploder shall be in accordance with Clause No. **6.0**.

4.9 In the case of a battery-operated e-Exploders, it shall be possible to either charge or replace the battery without disturbing or damaging components or circuit connections and without allowing the internal temperature to raise beyond 60 degrees centigrade. Replacement or charging of battery shall not be done belowground.

4.10 The e-Exploder shall provide suitable display and shall have an inbuilt feature such that the output terminals will become live only when the battery charge level is sufficient to complete the planned firing operation. There shall be visible indication of the battery charge level and shall have a built-in feature of auto shut off in the event the battery is unable to complete the firing operation unless charged.

4.11 The permitted type of e-Exploder shall also meet the following requirements:

Batch Testing of e-Exploder: Prior to use in mines, the manufacturer shall ensure that the exploder are subjected to the batch testing which shall include test of peak open circuit voltage across output terminals of the e-Exploder, residual charge which should be zero and duration of current impulse to the line which should be within four milliseconds of the operation of the firing command/switch. The manufacturer and users shall ensure that the samples of e-Exploders are drawn on random basis and consignment of the samples sealed jointly by the manufacturer and users. The number of samples to be drawn shall be @ of 1% of the total number of the procured exploders or a minimum of one e-exploder whichever is more and tested at Government approved laboratories / test houses accredited by NABL **[subject to confirmation of its accreditation granted by NABL for the relevant standard and that test house is not a part of any manufacturer's testing facility]**. The test house, upon receipt of the sealed sample(s) shall perform the requisite tests mentioned above and send original copies of the test reports both to manufacturer and the users which shall be maintained by them. In case, any of the samples fails in the above tests, the whole batch shall be discarded and no e-exploder shall be put into use.

4.12 Electrical Connections: All internal electrical connections shall be mechanically secured or soldered or brazed such that the e-Exploder meets the requirements of drop test and impact test (see Clause No. **8.4** and **8.5**).

4.13 (i) Open-Circuit Voltage (Peak): The value of maximum or peak open circuit voltage across output terminals of the e-Exploder is under consideration. Till the maximum value is

stipulated in the standard, the e-Exploder shall be tested as per the value declared by the manufacturer.

4.13(ii) The short circuit current limit of the permitted type of e-Exploder shall not exceed 50 mA.

4.14 CONSTRUCTION AND GENERAL REQUIREMENTS OF LOGGER:

4.14(i): The logger shall be so constructed that all parts are of satisfactory strength for the service intended. The enclosure case shall be robust such that it meets the requirements of Clause No. **8.4** and **8.5**, splash proof and dust tight (degree of protection IP 54) in accordance with IS: IS/IEC 60529:2001.

4.14(ii): Carrying Strap/handle - A carrying strap/handle shall be fastened securely to the body of the Logger. The carrying strap shall be of adequate strength to withstand an additional 50 percent weight of the Logger for 15 minutes without failure or break.

4.14(iii): The output terminals shall be provided with suitable connecting devices for firm connections. It shall not be possible to remove the connecting device from its position. It shall be possible for easy and firm connection of the leading wires of the electronic detonator cable, terminated with a connector or otherwise.

4.14(iv): The electronic circuit of the logger shall be adequately insulated from its enclosure and shall meet the requirements given in Clause No. **8.1**

4.14(v): For field programmable permitted electronic detonators to be used in underground coal mines, underground metalliferous mines (gassy, where Regulation 142 of the Metalliferous Mines Regulations, 1961 is applicable) and other hazardous locations, the Logger shall be intrinsically safe conforming to IS: 60079-11-2023.

4.14(vi): The instrument shall be so designed and constructed as to be incapable of firing even a single electronic detonator in the circuit. The short circuit current limit of the logger shall not exceed 20 mA for permitted type of logger.

5.0. MATERIALS:

5.1 Materials - Materials not specifically designated shall be of good commercial quality.

6.0 PERFORMANCE REQUIREMENTS:

6.1 The performance of e-Exploders for successful firing of any particular number of shots connected in parallel would depend on the characteristics of electronic detonators used and the conditions under which the shots are fired.

6.2.1 The e-Exploder manufacturer shall indicate in the instruction manual the maximum bus line/trunk line resistance.

6.2.2 In the case of capacitor type e-Exploders (if applicable), the value of capacitance, the maximum charged voltage and the external resistance shall be specified in the instruction manual. In addition, the total available energy/ohm/ms from the e-Exploder may also be specified therein.

6.3 The compliance of the e-Exploder with the requirements of this standard shall be checked by firing the specified number of electronic detonators in a manner indicated in the instruction manual.

6.4 The result of System function test conducted as per the **Appendix A** shall give satisfactory results.



6.5 The e-Exploder of a particular make shall be compatible with and fire only the electronic detonators of the same make.

Note: The compatibility of e-Exploder for a particular electronic detonator only shall be checked and tested to ensure that it shall not initiate/fire the other electronic detonators of different makes. The compatibility test shall be carried out by selecting a minimum five different makes of electronic detonators or e-Exploders available in Indian market.

7. MARKING

7.1 Each e-Exploder shall be marked with the following information on its rating plate:

- a) Name of the manufacturer;
- b) Type of e-Exploder (Permitted/General type);
- c) Rated Bus/trunk line resistance;
- d) Current in amperes/Maximum output voltage at terminals;
- e) The rated number of electronic detonators, the e-Exploder would fire
- f) Maximum power
- g) Maximum capacitance/Maximum inductance of the load supported by the e-Exploder; and
- h) Any other marking indicating the approval from the statutory authority.

7.2 Each e-Exploder shall be provided with an instruction manual giving necessary operating instructions, illustrations, testing instructions, and safety precautions to ensure proper operation and maintenance of the e-Exploder.

7.3 The e-Exploder may also be marked with the ISI Certification Mark.

NOTE - The use of the ISI Certification Mark is governed by the provisions of the Indian Standards Institution (Certification Marks) Act and the Rules and Regulations made there under. The ISI mark on products covered by an Indian Standard conveys the assurance that they have been produced to comply with the requirements of that standard under a well-defined system of inspection, testing and quality control which is devised and supervised by ISI and operated by the producer. ISI marked products are also continuously checked by ISI for conformity to that standard as a further safeguard Details of conditions, under which a license for the use of the ISI Certification Mark may be granted to manufacturers or processors, may be obtained from the Indian Standards Institution.

TESTS:

8.0 Acceptance Tests and the Sampling Plan - The acceptance tests and the sampling plan for e-Exploders shall be as agreed between the manufacturer and the user.

8.1 Insulation Resistance Test - The e-Exploder shall be kept for 24 hours in the humidity cabinet maintained at $27 \pm 2^\circ\text{C}$ and relative humidity not less than 95 percent. The insulation resistance of the exploder after humidity treatment shall be measured by insulation tester with a DC voltage of approximately 500 volts, applied between the line parts and the metal enclosure. The measurement shall be made one minute after the application of voltage. The insulation resistance so measured shall be not less than 50 mega ohm.

8.2 Dielectric Strength Test - The e-Exploder shall be capable of withstanding the application of 250 voltage (ac) applied for one minute between mutually insulated circuits and external enclosure/casing. After the high voltage test, the exploder shall meet the requirements given in Clause No. 6.

8.3 Endurance Test- All e-Exploders shall be capable of withstanding a minimum of 5000 actuations (presses) at the rate of not less than 6 operations per minute, without any mechanical failure or degradation in its performance. The e-Exploders after the mechanical endurance test shall comply with the requirements given in Clause No. 6.



8.4 Drop Test - A completely assembled e-Exploder shall be held in any position and then dropped once from a height of 1 m on to a concrete floor. This test may be repeated for severe positions with separate e-Exploders as desired by testing authorities before the grant of approval.

8.4.1 After the drop test, the e-Exploder shall be checked for the requirements given in Clause No. **8.5.2**.

8.5 Impact Test - The mechanical strength of the **enclosure** of the e-Exploder shall be tested by the impact of falling body having a hardened steel hemisphere of 25 mm diameter on the impinging end. The mass of this falling body shall be 1 kg which shall be dropped vertically from a height of 2 m. The falling body shall be aimed at the weakest points of enclosure. The e-Exploder enclosure shall be subjected to this test when placed on a rigid base. After the test it shall satisfy the requirements given in Clause No. **8.5.2** and also the dust tightness requirements of **IP 54** in accordance with **IS/IEC 60529:2001**.

8.5.1 The strength of the transparent parts of the e-Exploder-enclosure shall be tested by the impact of a falling body having on the impinging end a hemisphere of 25 mm diameter in polyamide (nylon 6.6) rockwell hardness R 100. The mass of the falling body shall be 1 kg which shall be dropped vertically from a height of 1 m. The transparent parts are tested when the e-Exploder is placed on a rigid base.

8.5.2 The impact test and drop test (Clause No. **8.4**) shall not produce a condition which invalidates the type of protection of the enclosure of electrical apparatus. Any deformation caused by these tests shall not prevent the proper operation of the electrical apparatus, neither reduce the creepage and clearance distances nor the degree of protection of the enclosure. It is accepted that superficial damage, chipping to paint work, breakage of cooling fans or similar parts of the electrical apparatus and small items shall be ignored.

8.6 System Function Test:

The e-Exploder as a system shall comply with the requirements of this test to determine the proper operation of the e-Exploders when tested in accordance with the procedure laid down by the manufacturer. The test requirements are as given in the **Appendix-A**

8.7 Electromagnetic Compatibility and Interference Testing:

The e-Exploder as a system shall comply with the requirements of this test to determine the proper operation of the e-Exploders when tested in accordance with the procedure laid down by the manufacturer. The test requirements are as given in the **Appendix-B**

Note: i) No dummy detonator shall be initiated in the absence of firing command during the test.

ii) After the completion of the test, no dummy electronic detonator shall be initiated till the firing command is issued. Next, the required sequence of operations shall be completed and firing command shall be issued. After issuing the firing command, all the dummy detonators shall be initiated.

iii) A temporary loss of function in equipment is allowed, providing the loss is self-recoverable (i.e. the equipment recovers itself and is suitable for further operation after the completion of the test), or can be restored by operation of the controls of the equipment (i.e. the equipment has not undergone any damage, and can be restored by operating the equipment or powering it off and on).

8.8 Abort function Test:

The e-Exploder as a system shall comply with the requirements of this test to determine the proper operation of the e-Exploders when tested in accordance with the procedure laid down by the manufacturer. The test requirements are as given in the **Appendix-C**.

8.9 Test report and certificate with effect of Clause No. 4.13(ii) regarding short circuit current limit not exceeding 50Ma.

8.10 Test report and certificate with effect of Clause No. 4.7 regarding requirement of terminating the output energy pulse within four milliseconds of release of the firing command.



A-1 INTRODUCTION**SYSTEM FUNCTIONAL TEST
(Clause 8.6)**

The purpose of this test is to determine the proper operation and satisfactory firing of the electronic detonators by the e-Exploder when connected and operated as specified in the manual of the manufacturer.

A-2 TEST PIECES

One e-Exploder and the maximum number of detonators supported in a single blast, as declared by the manufacturer, along with the trunk line cable of the resistance and length as recommended by the manufacturer.

A-3 PROCEDURE

- a) Use a firing device with enough charge, operated from the maximum distance specified by the manufacturer, the quantity of detonators as described above, connected to the firing device with the trunk line as described above.
- b) Minimum 5 test shall be conducted with all having the satisfactory results.

A-4 TEST REPORT

The test report shall have the following information:

- a) The total number of electronic detonators connected
- b) Maximum no. of detonators supported by the e-Exploder for a single blast.
- c) The serial numbers and the delay times loaded.
- d) The distance to the blaster and the first electronic detonator in the group.
- e) The resistance of the trunk line cable
- f) Whether all the detonators were programmed with the serial number and the delay time.
- g) Whether there was any loss of connectivity in attendance check
- h) Whether there was any failure in charging up
- i) The results of the examinations and tests obtained.



ELECTROMAGNETIC COMPATIBILITY AND INTERFERENCE TESTING (Clause 8.7)

B-1 INTRODUCTION

The purpose of this test is to verify that electronic initiation systems is safe and reliable when subjected to electromagnetic radiation.

B-2 TEST PIECES

For the tests described below, 10 dummy electronic detonators shall be used with all types of equipment from the electronic initiation system. When performing tests the equipment shall be connected and in operation during the respective tests with the dummy electronic detonators. All types of equipment in electronic initiation system shall be tested.

B-3 APPARATUS

As specified in the referred documents.

B-4 PROCEDURE

For each test below, record any loss of function or any unintended initiation of dummy detonators. In case of blast equipment, record the proper firing of all dummy electronic detonators after giving the firing command.

B-5 ELECTRICAL FAST TRANSIENT/ BURST IMMUNITY TEST

Perform the electrical fast transient/burst immunity test in accordance with IS 14700 (Part 4/Sec4): 2018 (identical adoption of IEC 61000-4-4) with a peak voltage of:

- a) 4 kV on the power supply port and on the control, lines connected to mains power and
- b) 2 kV on/off, signal and data ports.

B-6 ELECTROSTATIC DISCHARGE IMMUNITY TEST

Perform the electrostatic discharge immunity test in accordance with IS 14700 (Part 4/Sec 4): 2018 (identical adoption of IEC 61000-4-4) with a voltage corresponding to level 4:

- a) air discharge: 15 kV;
- b) contact discharge: 8 kV.

B-7 SAFETY AGAINST UNINTENDED INITIATION

For the tests described below, apply the worst-case condition (or conditions) for the specific system as close to the real field conditions as possible. Also consider the worst-condition regarding choice of wire length and possible connection of electronic detonators to earth.

For each test below record any unintended initiation of the dummy detonators:

- 1) Perform immunity to conducted disturbances, induced by radio-frequency fields test in accordance with IS 14700 (Part 4/Sec 6): 2016 (identical adoption of IEC 61000-4-6) with a voltage of 30 V at frequencies from 150 kHz - 80 MHz.
- 2) Perform radiated, radio-frequency, electromagnetic field immunity test in accordance with IS 14700 (Part 4/Sec 3): 2018 (identical adoption of IEC 61000-4-3) with a field strength of 30 V/m at frequencies from 80 MHz – 1 GHz.
- 3) Perform radiated, radio-frequency, electromagnetic field immunity test in accordance with IS 14700 (Part 4/Sec 3): 2018 (identical adoption of IEC 61000-4-3) with a field strength of 10 V/m at frequencies from 1 GHz – 2 GHz.

B-8 FUNCTIONING DURING STRESS

For the tests described below apply the worst-case condition for the specific system as close to the



real field conditions as possible regarding choice of wire length and possible connection of detonators to earth.

For each test record any loss of function or unintended initiation.

1) Perform immunity to conducted disturbances, induced by radio-frequency fields test in accordance with IS 14700 (Part 4/Sec 6): 2016 (identical adoption of IEC 61000-4-6) with a voltage of 10 V at frequencies from 150 kHz - 80 MHz. Select 30 discrete frequencies evenly distributed in the frequency range each for which a complete initiation process including firing is carried out. Additionally perform firing after completion of the test (i.e. without stress applied).

2) Perform radiated, radio-frequency, electromagnetic field immunity test in accordance with IS 14700 (Part 4/Sec 3): 2018 (identical adoption of IEC 61000-4-3) with a field strength of 10 V/m at frequencies from 80 MHz – 1 GHz. Select 10 discrete frequencies evenly distributed in the frequency range each for which a complete initiation process including firing is carried out. Additionally perform firing after completion of the test (i.e. without stress applied).

3) Perform radiated, radio-frequency, electromagnetic field immunity test in accordance with IS 14700 (Part 4/Sec 3): 2018 (identical adoption of IEC 61000-4-3) with a field strength of 3 V/m at frequencies from 1 GHz – 2 GHz. Select 10 discrete frequencies evenly distributed in the frequency range each for which a complete initiation process including firing is carried out. Additionally perform firing after completion of the test (i.e. without stress applied).

B-9 RADIO FREQUENCY EMISSION TEST

Perform the radio frequency emission test in accordance with CISPR 11, class A (industrial environmental).

NOTE: The result of the test to be declared by the manufacturer based on any authorized laboratory.

B-10 TEST REPORT

The test reports shall have following information:

- a) The results of the examinations and tests obtained.
- b) Whether or not the equipment was able to meet all the requirements.
- c) Whether there was any loss of operation or unintended initiation.



ABORT FUNCTIONAL TEST (Clause 8.8)

C-1 INTRODUCTION

The e-Exploder conducts a series of operations before the detonators are blasted. Where in there can be scenarios, which require the blast sequences to be aborted in between. Thus, e-Exploder must have a built-in safe abort mechanism.

An e-Exploder unit must have a switch/button which can override all the on-going sequences and revert detonators to a safe state while also disabling the Firing functionality. The manufacturer shall also declare the time required (after aborting and bus line disconnection) when the detonators can be considered safe.

C-2 PROCEDURE

Below test shall be conducted with the maximum no. of detonator supported by the e- Exploder declared by the manufacturer.

1. Connect the detonators with the blaster.
2. Start blast Sequence and bring the samples to charged or armed state, where they can initiate on receiving the firing signal.
3. Wait for 1 minute to ensure that the detonators will not explode in the absence of the firing command.
4. Issue 'Abort' command after which the detonators shall disarm/discharge in the time specified by the manufacturer or within 30 seconds whichever is less and shall revert to disarm/defuse state.
[Manufacturer shall provide the procedure to verify that the left-over stored energy will be less than the no fire energy of the fuse head in the e-detonator].
5. The exploder shall indicate by way of confirmation on the screen that the aborted operation was successful and if failed in a particular electronic detonator the same shall be indicated with distinctive number for further action as deemed fit.

C-3 TEST REPORT

The test report shall have the following information:

- a) The total number of electronic detonators connected.
- b) The serial numbers and the delay times loaded.
- c) The distance from the blaster to the first electronic detonator in the group.
- d) The resistance of the trunk line cable
- e) Whether all the detonators were programmed with the serial number and the delay time.
- f) Whether there was any loss of connectivity in attendance check
- g) Whether there was any failure in charging up
- h) The results shall indicate that whether all the connected detonators have defused/disarmed.



भारतीय मानक

IS 19359 : 2026

Indian Standard

ई विस्फोटक — विशिष्ट

e-Exploders — Specification

ICS 29.260.20

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Price Group 7

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Apparatus for Explosive Atmospheres Sectional Committee had been approved by the Electrotechnical Division Council.

An electronic exploder (e-exploder) is a set of devices consisting of a logger, a blaster and a continuity tester as independent devices with distinct functions and with independent authorization to ensure safe use.

These devices shall be used exclusively for communication with electronic detonators. It shall operate using a specific communication method and protocol compatible only with a corresponding type or model of electronic detonator designed using the same system architecture. Accordingly, e-exploders and electronic detonators from different manufacturers shall not be considered interchangeable. An e-exploder supplied by one manufacturer shall not be capable of initiating electronic detonators supplied by another manufacturer, due to differences in communication systems.

The e-exploder shall be designed to be portable and suitable for handling by an authorized shotfirer for field or mine use. It shall be provided with a human-machine interface, such as a keyboard or other suitable input device, capable of supporting multiple operational functions. These functions shall include, but not be limited to, defining, programming, checking, and initiating firing sequences for a batch of electronic detonators. The construction of the e-exploder shall be suitable for application in both underground and open-cast mining environments.

The e-exploder shall enable operation of the blasting process from a safe distance. For use in underground mines, the e-exploder shall additionally be equipped with intrinsically safe features, duly approved by the Directorate General of Mines Safety (DGMS), Government of India.

The e-exploder shall incorporate essential functionalities to support various operational stages, including:

- a) Verification of the health status of electronic detonators at the preliminary stage;
- b) Loading of serial numbers and delay timings into individual detonators in the programming stage; and
- c) Verification and confirmation of electrical connectivity of each electronic detonator to the trunk line cable prior to blasting.

During the blasting phase, the e-exploder shall support the following operations:

- a) Checking the attendance of all connected detonators;
- b) Initiating charge loading, with optional verification;
- c) Issuing the final fire command; and
- d) Providing the capability to issue a defuse or abort command to address unforeseen circumstances.

The e-exploder shall be housed in a robust and portable case designed to endure the mechanical and environmental stresses encountered in mining conditions. It shall be equipped with a physical access control key for switching the device on and off, along with password-based authentication for execution of commands.

As the Government of India has prohibited the manufacture, possession, and import of electric detonators (classified as explosives under Class 6, Division 3, or UN Class 1, Division 1) across the country with effect from 1 April 2025, this standard is intended to supersede IS 9836 : 1981 'Specification for exploders'.

The composition of the Committee responsible for the formulation of this standard is given in [Annex D](#).

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (second revision)'. The number of significant places retained in the rounded off value should be same as that of the specified value in this standard.

Indian Standard

e-EXPLODERS — SPECIFICATION

1 SCOPE

This standard covers the requirements of e-exploders and portable blasting/logging instruments used for shot firing operations for activating electronic detonator(s) during mining and non-mining blasting operations.

2 REFERENCES

2.1 Statutory Provisions

- a) *Metalliferous Mines Regulations, 1961*

2.2 Standards

The standards given below contain provisions which, through reference in this text, constitute provision of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards:

IS No.	Title
IS 14700	Electromagnetic compatibility (EMC):
(Part 4)	Testing and measurement techniques:
(Sec 3) : 2023/ IEC 61000-4-3 : 2020	Radiated radio-frequency electromagnetic field immunity test (<i>second revision</i>)
(Sec 4) : 2018/ IEC 61000-4-4 : 2012	Electrical fast transient/burst immunity test (<i>second revision</i>)
(Sec 6) : 2016/ IEC 61000-4-6 : 2013	Immunity to conducted disturbances, induced by radio-frequency fields
IS/IEC 60529 : 2001	Degrees of protection provided by enclosures (IP code)
IS/IEC 60079-11 : 2023	Explosive atmospheres: Part 11 equipment protection by intrinsic safety "i" (<i>second revision</i>)

3 DEFINITIONS

3.1 Electronic Detonator — Detonator assembly

in which the time delay is achieved by means of an electronic chip activated by electronic signature signal/stimuli.

3.2 Electronic Exploder (e-Exploder) — The Electronic Exploder (e-Exploder) means a system comprising a set of devices which employs electrical energy and coded signals for the purposes of programming, communication, verification, activation, initiation, and blasting of electronic detonator(s), and shall include a logger, a blaster, and a continuity tester, each constituted as an independent unit performing distinct functions and provided with independent authorization mechanisms so as to ensure safe and controlled operation.

3.3 Programmer — Also called as logger, it is an instrument used to program the serial number and the required delay into an electronic detonator with a mechanism to verify the same along with the healthiness of the detonator(s).

3.4 Blaster — An instrument used to carry out the final operations in the blasting sequence involving verification of the connected detonators from the firing chamber, to arm or charge the detonators and to issue the fire/defuse command.

3.5 Continuity Tester — Also known as connection tester, it is an instrument to verify the parallel connection of each detonator to the trunk line. The continuity tester used in underground mines will additionally come with intrinsically safe features as approved by Directorate General of Mines Safety (DGMS), Government of India.

4 CLASSIFICATIONS

The e-exploders covered by this standard shall be either of the following two types:

- a) *Permitted Type* — A type which is approved by the statutory authority namely, directorate general of mines safety, for use in underground coal mines, underground metalliferous mines (gassy, where Regulation 142 of the *Metalliferous Mines Regulations, 1961* is applicable) and other hazardous locations; and
- b) *General or Non-Permitted Type* — A type which is used for mining operations in open places or civil work where the danger from flammable gases is not present.

5 CONSTRUCTION AND GENERAL REQUIREMENTS

5.1 The e-exploder shall be so constructed that all parts are of satisfactory strength for the service intended. The enclosure case shall be robust such that it meets the requirements of **9.4** and **9.5**, splash proof and dust tight (degree of protection IP 54) in accordance with IS/IEC 60529 as amended time to time.

The enclosure of the permitted type e-exploders shall be so constructed as not to give rise to ignition hazard of gas and coal dust due to frictional spark, static charges and leakage current.

The circuit of the e-exploder shall be intrinsically safe conforming to IS/IEC 60079-11, as amended time to time.

5.2 Carrying Strap/Handle/Case

A carrying strap/handle shall be fastened securely to the body of the e-exploder or its carrying case. The carrying strap shall be of adequate strength to withstand an additional 50 percent weight of e-exploder for 15 minutes without failure or break.

5.3 The output terminals shall be provided with suitable connecting devices for firm connections. It should not be possible to remove the connecting device from its position. It shall be possible for easy and firm connection of shot firing cable with the terminals.

5.4 The firing circuit of the e-exploder shall be adequately insulated from its enclosure and shall meet the requirements given in **9.1**.

5.5 It shall not be possible to keep the e-exploder devices in "ON" condition unless the key is inserted. The operating key shall be specific to each e-exploder to avoid the operations by unauthorised persons using a generally available key of similar design or construction.

5.6 The e-exploder (Blasting unit) shall be so constructed that:

- a) It shall be operated by a removable key followed by an authentication password for activation;
- b) The arming/fire command from the blaster of the e-exploder shall be by an additional and separate password enabled operation;
- c) No residual energy is left at the terminals after the completion of operation sequence; and

- d) The circuit of the blasting unit shall be intrinsically safe conforming to IS/IEC 60079-11, as amended time to time.

5.7 The permitted type e-exploders shall have suitable devices to terminate the output energy pulse within four milliseconds of the release of the firing command.

5.8 Each e-exploder shall be capable of firing a prescribed number of electronic detonators in the manner recommended by the manufacturer of the e-exploder. The performance requirements of the e-exploder shall be in accordance with **7**.

5.9 In the case of a battery-operated e-exploders, it shall be possible to either charge or replace the battery without disturbing or damaging components or circuit connections and without allowing the internal temperature to raise beyond 60 °C. Replacement or charging of battery shall not be done belowground.

5.10 The e-exploder shall provide suitable display and shall have an inbuilt feature such that the output terminals will become live only when the battery charge level is sufficient to complete the planned firing operation. There shall be visible indication of the battery charge level and shall have a built-in feature of auto shut off in the event the battery is unable to complete the firing operation unless charged.

5.11 Batch Testing of e-Exploder

Prior to use in mines, the manufacturer shall ensure that the exploder is subjected to the batch testing which shall include test of peak open circuit voltage across output terminals of the e-exploder, residual charge which should be zero and duration of current impulse to the line which should be within four milliseconds of the operation of the firing command/switch. The manufacturer and users shall ensure that the samples of e-exploders are drawn on random basis and consignment of the samples sealed jointly by the manufacturer and users. The number of samples to be drawn shall be 1 percent of the total number of the procured exploders or a minimum of one e-exploder whichever is more and tested at government approved laboratories/test houses. The laboratory shall be accredited to laboratory quality management system as per IS/ISO/IEC 17025 (latest implemented/concurrent version). The accreditation body, through which the laboratory is accredited, shall be a full member of International laboratory accreditation cooperation (ILAC) and/or Asia Pacific Accreditation Co-operation (APAC) or any other regional accreditation cooperation body. The test house, upon receipt of the sealed sample(s) shall perform the requisite tests mentioned above and

send original copies of the test reports both to manufacturer and the users which shall be maintained by them. In case, any of the samples fails in the above tests, the whole batch shall be discarded and no e-exploder shall be put into use.

5.12 Electrical Connections

All internal electrical connections shall be mechanically secured or soldered or brazed such that the e-exploder meets the requirements of drop test and impact test (see [9.5](#) and [9.6](#)).

5.13 The following are the voltage and current requirement of e-exploder:

- a) *Open-Circuit Voltage (Peak)* — The value of maximum or peak open circuit voltage across output terminals of the e-exploder is under consideration. Till the maximum value is stipulated in the standard, the e-exploder shall be tested as per the value declared by the manufacturer; and
- b) The short circuit current limit of the permitted type of e-exploder shall not exceed 50 mA.

5.14 Construction and General Requirements of Logger

- a) The logger shall be so constructed that all parts are of satisfactory strength for the service intended. The enclosure case shall be robust such that it meets the requirements of [9.4](#) and [9.5](#), splash proof and dust tight (degree of protection IP 54) in accordance with IS/IEC 60529;
- b) *Carrying Strap/Handle* — A carrying strap/handle shall be fastened securely to the body of the Logger. The carrying strap shall be of adequate strength to withstand an additional 50 percent weight of the Logger for 15 minutes without failure or break;
- c) The output terminals shall be provided with suitable connecting devices for firm connections. It shall not be possible to remove the connecting device from its position. It shall be possible for easy and firm connection of the leading wires of the electronic detonator cable, terminated with a connector or otherwise;
- d) The electronic circuit of the logger shall be adequately insulated from its enclosure and shall meet the requirements given in [9.1](#);
- e) For field programmable permitted electronic detonators to be used in underground coal mines, underground Metalliferous mines (gassy, where regulation 142 of the Metalliferous Mines Regulations, 1961 is applicable) and other

hazardous locations, the logger shall be intrinsically safe conforming to IS/IEC 60079-11; and

- f) The instrument shall be designed and constructed in such a manner that it is incapable of initiating even a single electronic detonator within the circuit under any condition. Furthermore, the short-circuit current of the logger shall be limited to a short-circuit current of 20 mA for the permitted type of logger.

6 MATERIALS

Materials not specifically designated shall be of good commercial quality.

7 PERFORMANCE REQUIREMENTS

7.1 The performance of e-exploders for successful firing of any particular number of shots connected in parallel would depend on the characteristics of electronic detonators used and the conditions under which the shots are fired.

7.2 The manufacturer of the e-exploder shall specify, in the instruction manual, the maximum permissible resistance of the bus line/trunk line, expressed in ohms (Ω). In the case of capacitor-type e-exploders, where applicable, the capacitance shall be specified in microfarads (μF) or picofarads (pF), as the case may be, along with the maximum charged voltage expressed in volts (V) or millivolts (mV), as applicable, and the corresponding external resistance in ohms (Ω).

In addition, the total available output energy shall be indicated in millijoules (mJ), along with the effective discharge characteristics in relation to circuit resistance (Ω) and time duration (milliseconds), as applicable.

7.3 The compliance of the e-exploder with the requirements of this standard shall be checked by firing the specified number of electronic detonators in a manner indicated in the instruction manual.

7.4 The result of system function test conducted as per the [Annex A](#) shall give satisfactory results.

7.5 The e-exploder of a particular make shall be compatible with and fire only the electronic detonators of the same make.

NOTE — The compatibility of e-exploder for a particular electronic detonator only shall be checked and tested to ensure that it shall not initiate/fire the other electronic detonators of different makes. The compatibility test shall be carried out by selecting a minimum five different makes of electronic detonators or e-exploders available in Indian market.

8 MARKING

8.1 Each e-exploder shall be marked with the

following information on its rating plate:

- a) Name of the manufacturer;
- b) Type of e-exploder (permitted/general type);
- c) Type of ex protection of equipment;
- d) Batch number;
- e) Rated bus/trunk line resistance;
- f) Current in amperes/maximum output voltage at terminals;
- g) The rated number of electronic detonators, the e-exploder would fire;
- h) Maximum power in Watts or mW;
- j) Maximum capacitance/maximum inductance of the load supported by the e-exploder;
- k) IP rating;
- m) Ambient temperature that is, T ambient;
- n) T class;
- p) country of origin and month/year of manufacture;
- q) Altitude up to which the e-exploder is effective;
- r) limits of storage temperature and ambient operating temperature may be specified (not to exceed -10°C to $+55^{\circ}\text{C}$ both); and
- s) Any other marking indicating the approval from the statutory authority.

8.2 Each e-exploder shall be provided with an instruction manual giving necessary operating instructions, illustrations, testing instructions, and safety precautions to ensure proper operation and maintenance of the e-exploder.

8.3 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the Bureau of Indian Standards Act, 2016 and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

9 TESTS

9.1 Acceptance Tests and the Sampling Plan

The acceptance tests and the sampling plan for e-exploders shall be as agreed between the manufacturer and the user.

9.2 Insulation Resistance Test

The e-exploder shall be kept for 24 hours in the humidity cabinet maintained at $27^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and relative humidity not less than 95 percent. The insulation resistance of the exploder after humidity treatment shall be measured by insulation tester with a dc voltage of approximately 500 volts, applied between the line parts and the metal enclosure.

The measurement shall be made one minute after the application of voltage. The insulation resistance so measured shall be not less than 50 mega ohms.

9.3 Dielectric Strength Test

The e-exploder shall be capable of withstanding the application of 250 voltage (a.c.) applied for one minute between mutually insulated circuits and external enclosure/casing. After the high voltage test, the exploder shall meet the requirements given in 7.

9.4 Endurance Test

All e-exploders shall be capable of withstanding a minimum of 5 000 actuations (presses) at the rate of not less than 6 operations per minute, without any mechanical failure or degradation in its performance. The e-exploders after the mechanical endurance test shall comply with the requirements given in 7.

9.5 Drop Test

A completely assembled e-exploder shall be held in any position and then dropped once from a height of 1 m on to a concrete floor. This test may be repeated for severe positions with separate e-exploders as desired by testing authorities before the grant of approval.

9.5.1 After the drop test, the e-exploder shall be checked for the requirements given in 9.6.2.

9.6 Impact Test

The mechanical strength of the enclosure of the e-exploder shall be tested by the impact of falling body having a hardened steel hemisphere of 25 mm diameter on the impinging end. The mass of this falling body shall be 1 kg which shall be dropped vertically from a height of 2 m.

The falling body shall be aimed at the weakest points of enclosure. The e-exploder enclosure shall be subjected to this test when placed on a rigid base. After the test it shall satisfy the requirements given in 9.6.2 and also the dust tightness requirements of IP 54 in accordance with IS/IEC 60529.

9.6.1 The strength of the transparent parts of the e-exploder enclosure shall be tested by the impact of a falling body having on the impinging end a hemisphere of 25 mm diameter in polyamide (nylon 6.6) rockwell hardness R 100. The mass of the falling body shall be 1 kg which shall be dropped vertically from a height of 1 m. The transparent parts are tested when the e-exploder is placed on a rigid base.

9.6.2 The drop test and impact test (*see* 9.5 and 9.6) shall not produce a condition which invalidates the type of protection of the enclosure of electrical apparatus. Any deformation caused by these tests shall not prevent the proper operation of the electrical apparatus, nor reduce the creepage and clearance distances nor the degree of protection of the enclosure. It is accepted that superficial damage, chipping to paint work, breakage of cooling fans or similar parts of the electrical apparatus and small items shall be ignored.

9.7 System Function Test

The e-exploder as a system shall comply with the requirements of this test to determine the proper operation of the e-exploders when tested in accordance with the procedure laid down by the manufacturer. The test requirements are as given in the [Annex A](#).

9.8 Electromagnetic Compatibility and Interference Testing

The e-exploder as a system shall comply with the

requirements of this test to determine the proper operation of the e-exploders when tested in accordance with the procedure laid down by the manufacturer. The test requirements are as given in the [Annex B](#).

NOTES

1 No dummy detonator shall be initiated in the absence of firing command during the test.

2 After the completion of the test, no dummy electronic detonator shall be initiated till the firing command is issued. Next, the required sequence of operations shall be completed and firing command shall be issued. After issuing the firing command, all the dummy detonators shall be initiated.

3 A temporary loss of function in equipment is allowed, providing the loss is self-recoverable (the equipment recovers itself and is suitable for further operation after the completion of the test), or can be restored by operation of the controls of the equipment (the equipment has not undergone any damage, and can be restored by operating the equipment or powering it off and on).

9.9 Abort Function Test

The e-exploder as a system shall comply with the requirements of this test to determine the proper operation of the e-exploders when tested in accordance with the procedure laid down by the manufacturer. The test requirements are as given in the [Annex C](#).

ANNEX A

(Clauses 7.4 and 9.7)

SYSTEM FUNCTION TEST

A-1 INTRODUCTION

The purpose of this test is to determine the proper operation and satisfactory firing of the electronic detonators by the e-exploder when connected and operated as specified in the manual of the manufacturer.

A-2 TEST PIECES

One e-exploder and the maximum number of detonators supported in a single blast, as declared by the manufacturer, along with the trunk line cable of the resistance and length as recommended by the manufacturer.

A-3 PROCEDURE

- a) Use a firing device with enough charge, operated from the maximum distance specified by the manufacturer, the quantity of detonators as described above, connected to the firing device with the trunk line as described above; and
- b) Minimum 5 test shall be conducted with all having the satisfactory results.

A-4 TEST REPORT

The test report shall have the following information:

- a) The total number of electronic detonators connected;
- b) Maximum no. of detonators supported by the e-exploder for a single blast;
- c) The serial numbers and the delay times loaded;
- d) The distance to the blaster and the first electronic detonator in the group;
- e) The resistance of the trunk line cable;
- f) Whether all the detonators were programmed with the serial number and the delay time;
- g) Whether there was any loss of connectivity in attendance check;
- h) Whether there was any failure in charging up; and
- j) The results of the examinations and tests obtained.

ANNEX B

(Clause 9.8)

ELECTROMAGNETIC COMPATIBILITY AND INTERFERENCE TESTING

B-1 INTRODUCTION

The purpose of this test is to verify that electronic initiation systems is safe and reliable when subjected to electromagnetic radiation.

B-2 TEST PIECES

For the tests described below, 10 dummy electronic detonators shall be used with all types of equipment from the electronic initiation system. When performing tests, the equipment shall be connected and in operation during the respective tests with the dummy electronic detonators. All types of equipment in electronic initiation system shall be tested.

B-3 APPARATUS

As specified in the referred documents.

B-4 PROCEDURE

For each test below, record any loss of function or any unintended initiation of dummy detonators.

In case of blast equipment, record the proper firing of all dummy electronic detonators after giving the firing command.

B-5 ELECTRICAL FAST TRANSIENT/BURST IMMUNITY TEST

Perform the electrical fast transient/burst immunity test in accordance with IS 14700 (Part 4/Sec 4) with a peak voltage of:

- a) 4 kV on the power supply port and on the control lines connected to mains power; and
- b) 2 kV on/off, signal and data ports.

B-6 ELECTROSTATIC DISCHARGE IMMUNITY TEST

Perform the electrostatic discharge immunity test in accordance with IS 14700 (Part 4/Sec 4) with a voltage corresponding to level 4:

- a) Air discharge: 15 kV; and
- b) Contact discharge: 8 kV.

B-7 SAFETY AGAINST UNINTENDED INITIATION

For the tests described below, apply the worst-case condition (or conditions) for the specific system as close to the real field conditions as possible. Also consider the worst-condition regarding choice of wire length and possible connection of electronic detonators to earth.

For each test below record any unintended initiation of the dummy detonators:

- a) Perform immunity to conducted disturbances, induced by radio-frequency fields test in accordance with IS 14700 (Part 4/Sec 6) with a voltage of 30 V at frequencies from 150 kHz to 80 MHz;
- b) Perform radiated, radio-frequency, electromagnetic field immunity test in accordance with IS 14700 (Part 4/Sec 3) with a field strength of 30 V/m at frequencies from 80 MHz to 1 GHz;
- c) Perform radiated, radio-frequency, electromagnetic field immunity test in accordance with IS 14700 (Part 4/Sec 3) with a field strength of 10 V/m at frequencies from 1 GHz to 2 GHz.

B-8 FUNCTIONING DURING STRESS

For the tests described below apply the worst-case condition for the specific system as close to the real field conditions as possible regarding choice of wire length and possible connection of detonators to earth.

For each test record any loss of function or unintended initiation.

- a) Perform immunity to conducted disturbances, induced by radio-frequency fields test in accordance with IS 14700 (Part 4/Sec 6) with a voltage of 10 V at frequencies from 150 kHz to 80 MHz. Select 30 discrete frequencies evenly distributed in the frequency range each for which a complete initiation process including firing is carried out.

Additionally, perform firing after completion of the test (without stress applied).

- b) Perform radiated, radio-frequency, electromagnetic field immunity test in accordance with IS 14700 (Part 4/Sec 3) with a field strength of 10 V/m at frequencies from 80 MHz – 1 GHz. Select 10 discrete frequencies evenly distributed in the frequency range each for which a complete initiation process including firing is carried out.

Additionally, perform firing after completion of the test (without stress applied).

- c) Perform radiated, radio-frequency, electromagnetic field immunity test in accordance with IS 14700 (Part 4/Sec 3) with a field strength of 3 V/m at frequencies from 1 GHz to 2 GHz. Select 10 discrete frequencies evenly distributed in the frequency range each for which a complete initiation process including firing is carried out.

Additionally, perform firing after completion of the test (without stress applied).

B-9 RADIO FREQUENCY EMISSION TEST

Perform the radio frequency emission test in accordance with CISPR 11, class A (industrial environmental).

NOTE — The result of the test to be declared by the manufacturer based on any authorized laboratory.

B-10 TEST REPORT

The test reports shall have following information:

- a) The results of the examinations and tests obtained;
- b) Whether or not the equipment was able to meet all the requirements; and
- c) Whether there was any loss of operation or unintended initiation.

ANNEX C

(Clause 9.9)

ABORT FUNCTION TEST

C-1 INTRODUCTION

The e-exploder conducts a series of operations before the detonators are blasted. Where in there can be scenarios, which require the blast sequences to be aborted in between. Thus, e-exploder must have a built-in safe abort mechanism.

An e-exploder unit must have a switch/button which can override all the on-going sequences and revert detonators to a safe state while also disabling the firing functionality. The manufacturer shall also declare the time required (after aborting and bus line disconnection) when the detonators can be considered safe.

C-2 PROCEDURE

Below test shall be conducted with the maximum number of detonator supported by the e-exploder declared by the manufacturer:

- a) Connect the detonators with the blaster;
- b) Start blast sequence and bring the samples to charged or armed state, where they can initiate on receiving the firing signal;
- c) Wait for 1 minute to ensure that the detonators will not explode in the absence of the firing command;
- d) Issue 'Abort' command after which the detonators shall disarm/discharge in the time specified by the manufacturer or within 30 seconds whichever is less and shall revert to disarm/defuse state;

NOTE — Manufacturer shall provide the procedure to verify that the left over stored energy will be less than the no fire energy of the fuse head in the electric detonator.

- e) The exploder shall indicate by way of confirmation on the screen that the aborted operation was successful and if failed in a particular electronic detonator the same shall be indicated with distinctive number for further action as deemed fit.

C-3 TEST REPORT

The test report shall have the following information:

- a) The total number of electronic detonators connected;
- b) The serial numbers and the delay times loaded;
- c) The distance from the blaster to the first electronic detonator in the group;
- d) The resistance of the trunk line cable;
- e) Whether all the detonators were programmed with the serial number and the delay time;
- f) Whether there was any loss of connectivity in attendance check;
- g) Whether there was any failure in charging up; and
- h) The results shall indicate that whether all the connected detonators have defused/disarmed.

ANNEX D

(Foreword)

COMMITTEE COMPOSITION

Electrical Apparatus for Explosive Atmospheres Sectional Committee, ETD 22

<i>Organization</i>	<i>Representative(s)</i>
Directorate General of Mines Safety, Dhanbad	SHRI AJAY SINGH (<i>Chairperson</i>)
Baliga Lighting Equipments Private Limited, Chennai	SHRI B. RAMESH BALIGA SHRI K. SAKTHIVAL (<i>Alternate</i>)
Bharat Bijlee Limited, Mumbai	SHRI SALIL KUMAR SHRI JAYENDRA GARUD (<i>Alternate</i>)
Bharat Heavy Electrical Limited, New Delhi	SHRI MUKESH KUMAR MARAVI SHRI YATHARTH SETH (<i>Alternate</i>)
Bharat Petroleum Corporation Limited, Mumbai	SHRI KAUSTUBH PRASAD SHRI VEERUBHOTLA CHANDRA SHEKHAR (<i>Alternate</i>)
CG Power and Industrial Solutions, Mumbai	SHRI SANDEEP R. BACHKAR SHRI PRAVIN KUMAR PANDIT (<i>Alternate</i>)
Coal India Limited, Kolkata	SHRI SRI NARAYAN DASH SHRI SRI P. K. MANDAL (<i>Alternate</i>)
Confederation of Indian Industry, New Delhi	SHRI K. NANDAKUMAR
CSIR - Central Institute for Mining and Fuel Research, Dhanbad	SHRI AWANINDRA PRATAP SINGH SHRI RAKESH KUMAR MISHRA (<i>Alternate</i>)
Defence Research Development Organization, Ministry of Defence, New Delhi	SHRI ANIL KUMAR MS MEETA BAGHEL (<i>Alternate</i>)
Directorate General of Mines Safety, Dhanbad	SHRI RAJ KUMAR
Electrical Research and Development Association, Vadodara	SHRI RAVI SINGH SHRI JITENDRA TAHILWANI (<i>Alternate</i>)
EMERSON, Pune	SHRI SUMIT CHOPADA SHRI AJAY PATIDAR (<i>Alternate</i>)
Engineers India Limited, New Delhi	SHRI HARISH KUMAR MS RASHMI SINGH RATHAUR (<i>Alternate I</i>) SHRI SUBHADIP ACHARYA (<i>Alternate II</i>)
ExSafeT Services Group, Vadodara	SHRI K. M. SINGH
Flameproof Equipments Private Limite, Mumbai	SHRI HARESH DHOOT
Gujarat Narmada Valley Fertilizers and Chemicals Limited, Bharuch	SHRI BHATT KETAN KUNDANLAL SHRI M. U. THAKKER (<i>Alternate</i>)
Indian Flameproof Manufacturers Association, Mumbai	SHRI R. H. PATEL SHRI NILESH AGARWAL (<i>Alternate</i>)
Indian Institute of Technology (ISM), Dhanbad	SHRI TANMOY MAITY SHRI ANAND SHANKAR HATI (<i>Alternate</i>)

<i>Organization</i>	<i>Representative(s)</i>
Indian Oil Corporation Limited, New Delhi	MS NEETA AGARWAL SHRI SRI RAMAVTAR MEENA (<i>Alternate</i>)
Intertek India Private Limited, Gurugram	SHRI RAKESH CHAURASIA SHRI IRFAN KHAN (<i>Alternate</i>)
Karandikr Laboratories Private Limited, Mumbai	SHRI AJIT VASUDEV KARANDIKAR SHRI RAVINDRA KESHAV PARANJPE (<i>Alternate</i>)
Mineline Private Limited, Kolkata	MS NANDINI CHAKRAVARTY SHRI TAPAS MAZUMDAR (<i>Alternate</i>)
Oil and Natural Gas Corporation Limited, New Delhi	SHRI MUKESH KUMAR SHRI SREE HARI SANKAR TV (<i>Alternate</i>)
Oil Industry Safety Directorate, Noida	SHRI MURARI MOHAN PRASAD SHRI NAWAL PANDE (<i>Alternate</i>)
R Stahl Private Limited, Chennai	SHRI K. ARUNACHALAM SHRI D. SATISH KUMAR (<i>Alternate</i>)
Reliance India Limited, Mumbai	SHRI M. N. BHAGAVAN
The Singareni Collieries Company Limited, Kothagudem	SHRI NETI NAGESWARA RAO SHRI Y. RAGHURAMI REDDY (<i>Alternate</i>)
In Personal Capacity (<i>D 45 Anand Niketan New Delhi – 110021</i>)	SHRI NIRAJ SETHI
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